1.15 Graph Sketching: Cubic Graphs

Worked Solutions

1a)

$$y = x^3 - 4$$

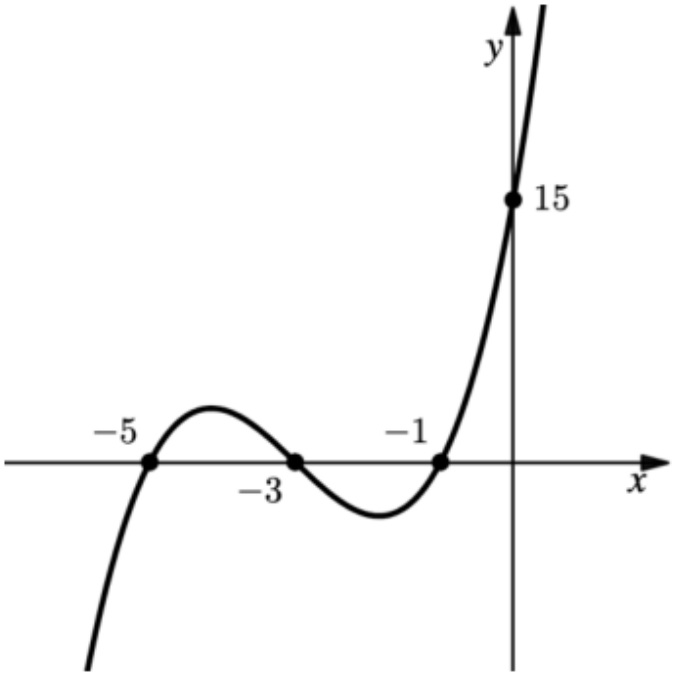
1b)

$$y = (x-2)^3 + 1 \Rightarrow y = x^3 - 6x^2 + 12x - 7$$

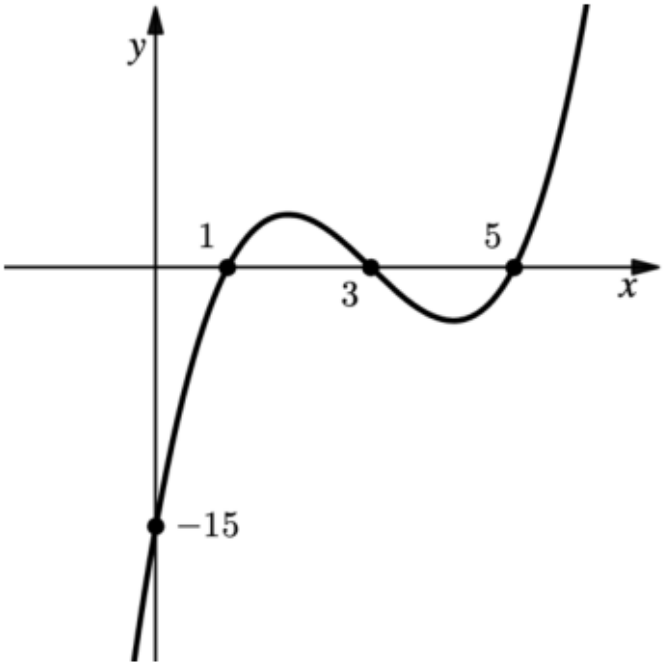
1c)

$$y = (x+3)^3 - 20 \Rightarrow y = x^3 + 9x^2 + 27x + 7$$

2a)



2b)



3a)

$$y = -3 - x^3 \Rightarrow y = -x^3 - 3$$

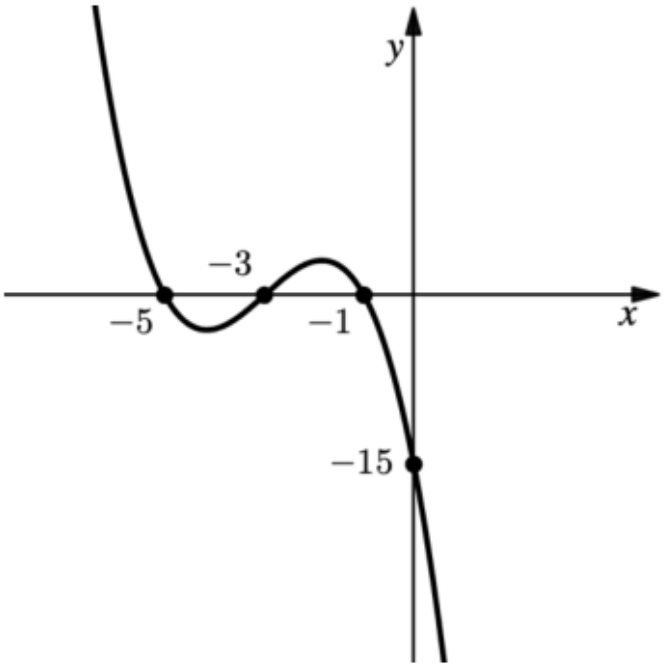
3b)

$$y = -(x+4)^3 \Rightarrow y = -x^3 - 12x^2 - 48x - 64$$

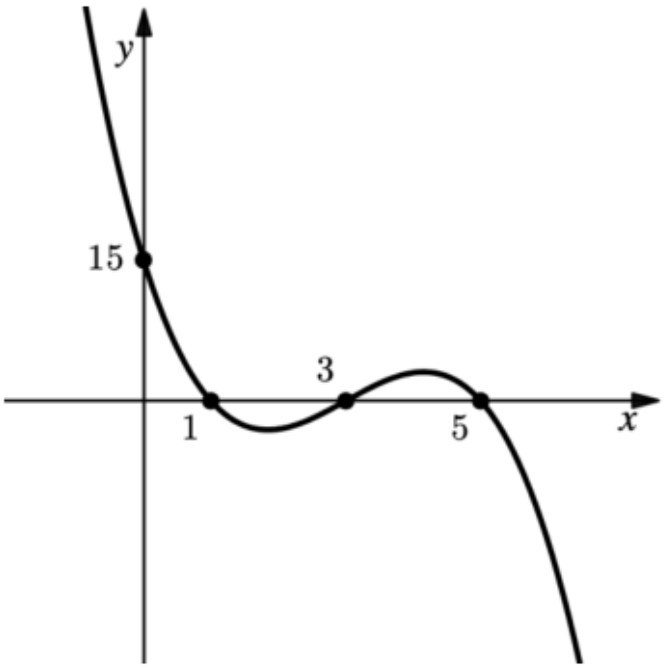
3c)

$$\begin{aligned} y &= 20 - (x-2)^3 \\ \Rightarrow y &= 20 - (x^3 - 6x^2 + 12x - 8) \\ \Rightarrow y &= -x^3 + 6x^2 - 12x + 28 \end{aligned}$$

4a)



4b)



5a)

$$y = (x+3)(x+1)(x-2)$$

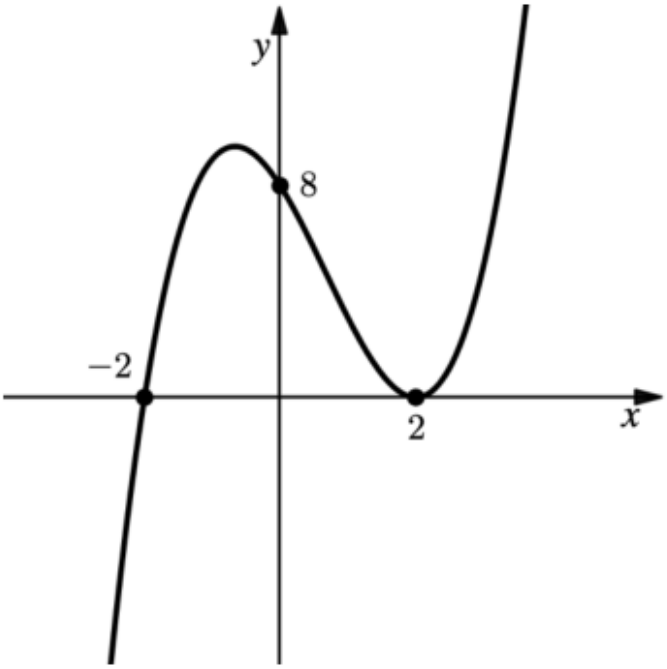
5b)

$$y = (x + 2)(x - 1)(x - 4)$$

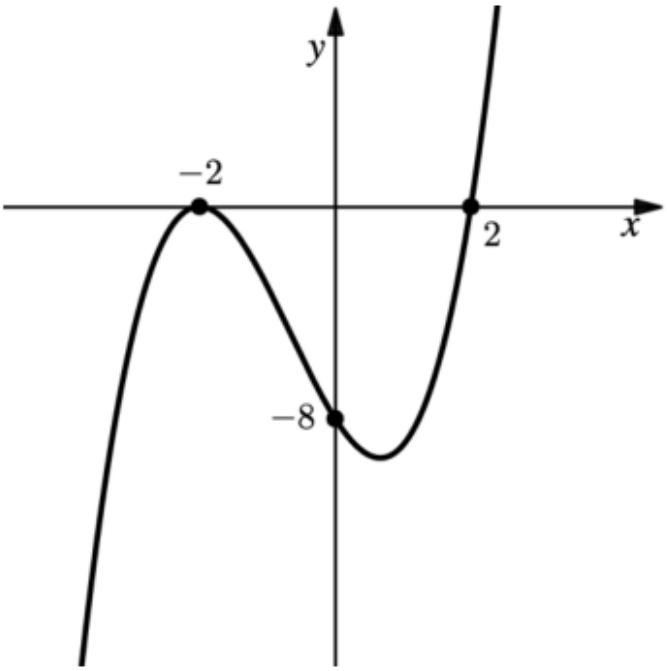
5c)

$$y = (x - 2)(x - 3)(x - 5)$$

6a)



6b)



7a)

$$y = (x+4)(x+2)^2$$

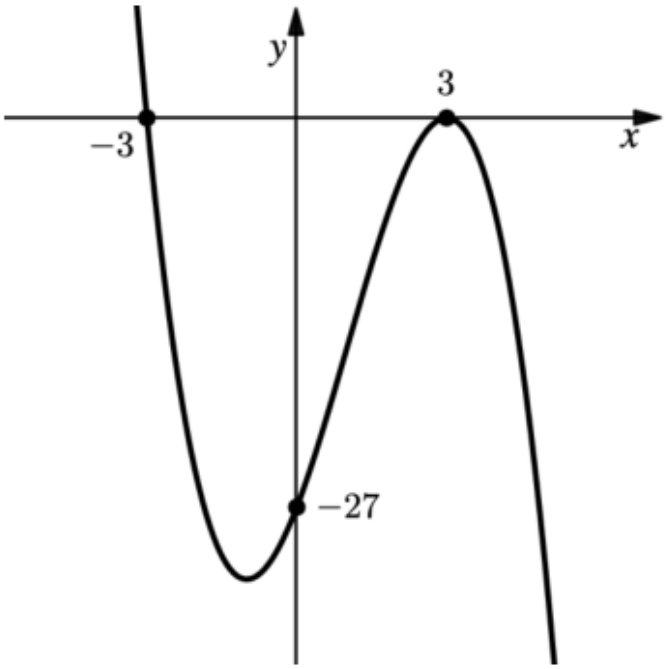
7b)

$$y = (x+3)^2(x-1)$$

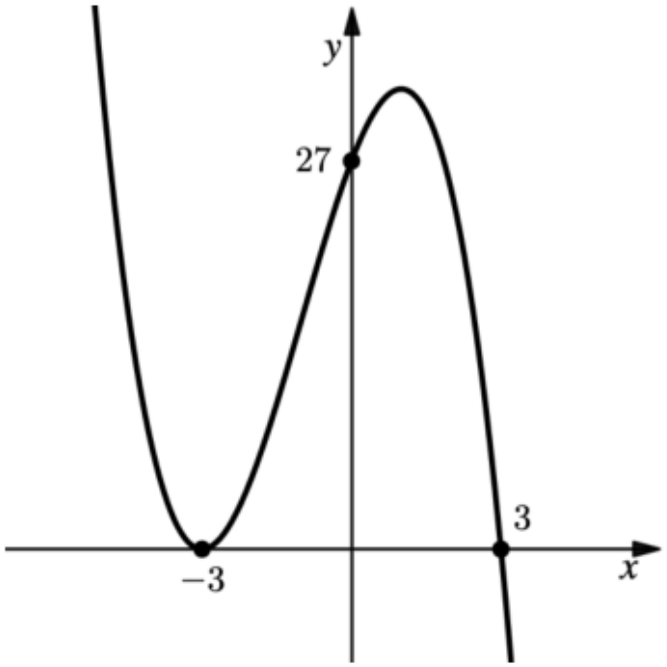
7c)

$$y = (x+4)x^2$$

8a)



8b)



9a)

$$y = k(x+3)(x-2)(x-4)$$
$$(0,-48): \quad -48 = k(3)(-2)(-4) \Rightarrow k = -2$$
$$y = -2(x+3)(x-2)(x-4)$$

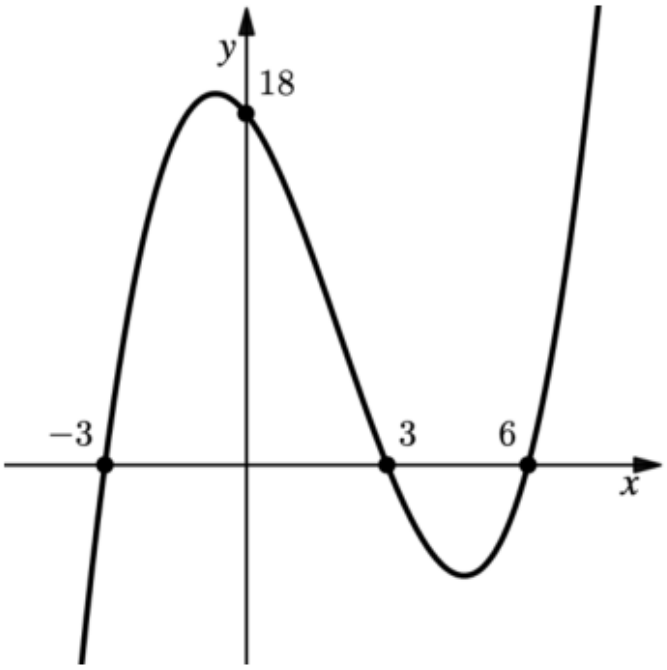
9b)

$$\begin{aligned} y &= k(x+6)(x+3)(x-2) \\ (0,24): \ 24 &= k(6)(3)(-2) \Rightarrow k = -\tfrac{2}{3} \\ y &= -\tfrac{2}{3}(x+6)(x+3)(x-2) \end{aligned}$$

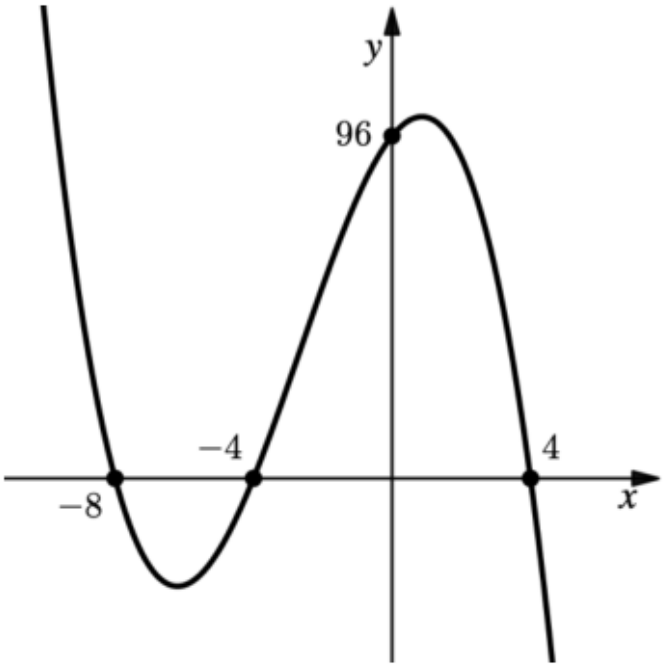
9c)

$$\begin{aligned} y &= kx(x+2)(x-4) \\ (2,-8): \ -8 &= k(2)(4)(-2) \Rightarrow k = \tfrac{1}{2} \\ y &= \tfrac{1}{2}x(x+2)(x-4) \end{aligned}$$

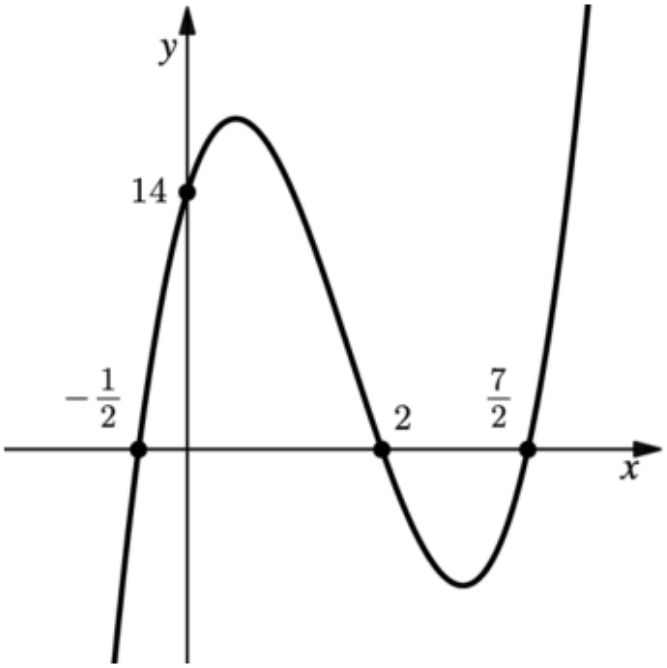
10a)



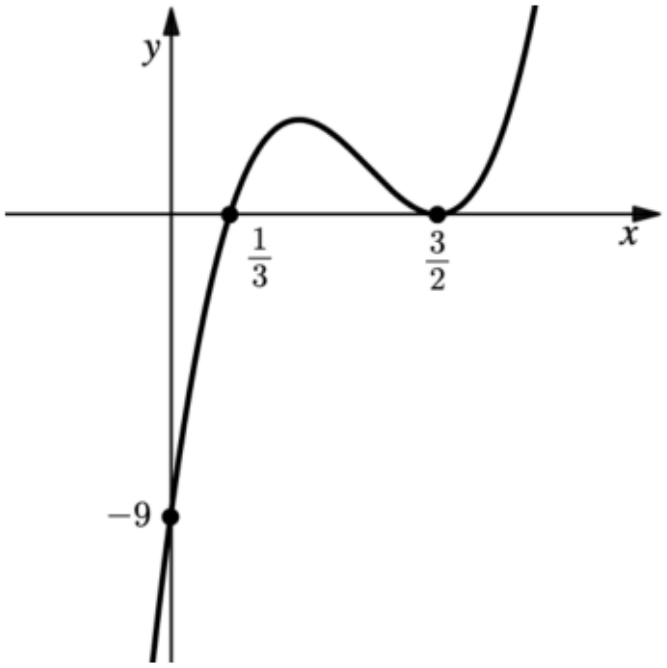
10b)



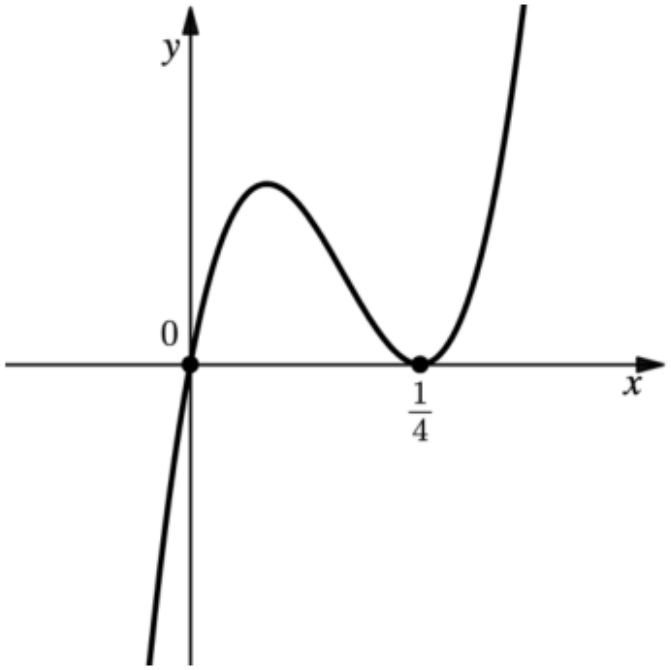
11a)



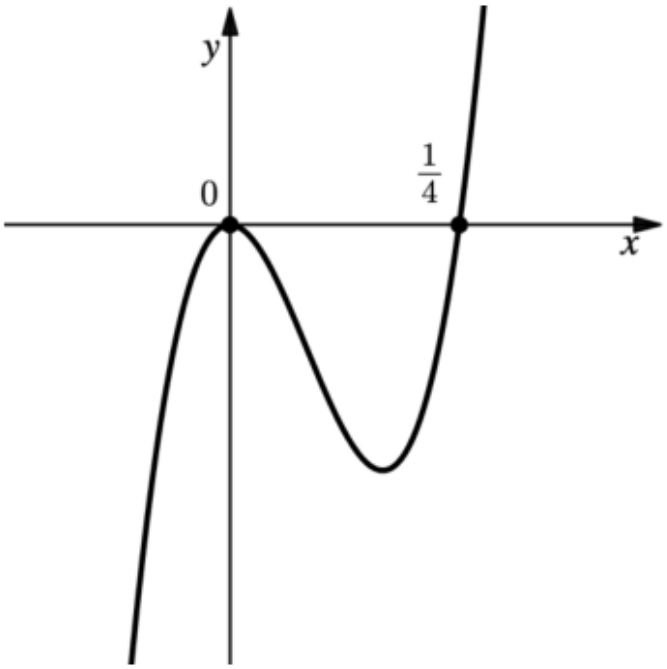
11b)



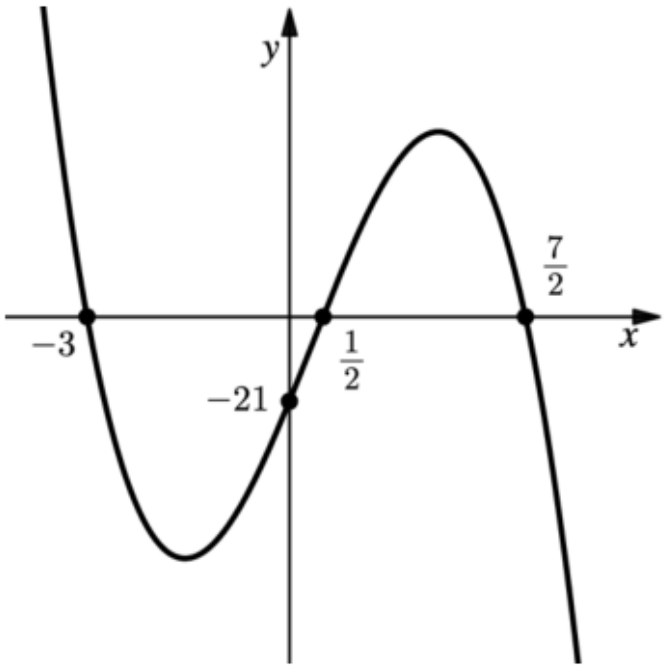
11c)



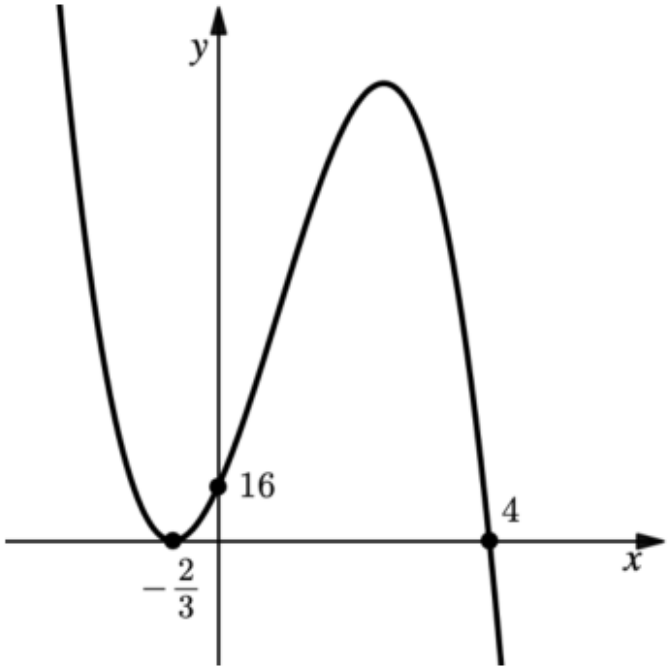
11d)



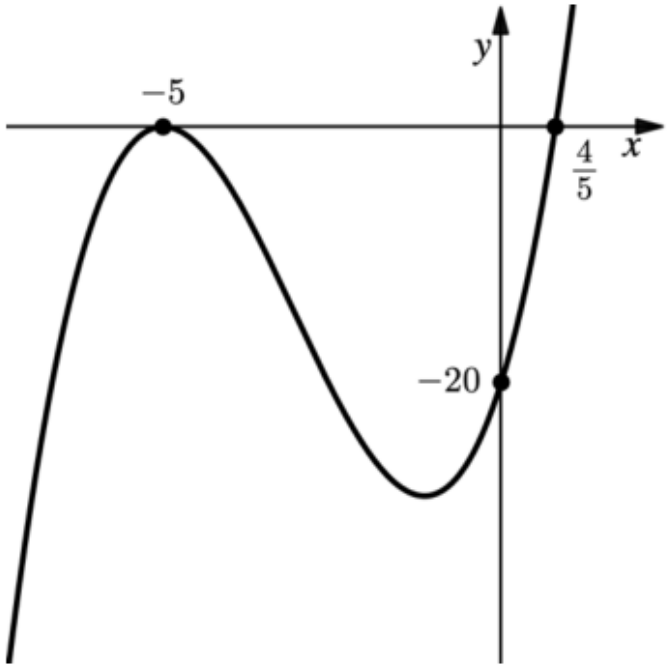
12a)



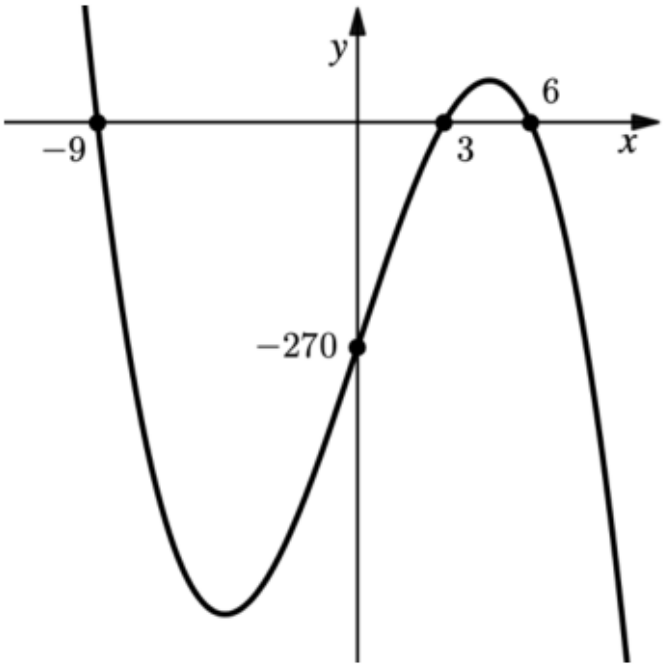
12b)



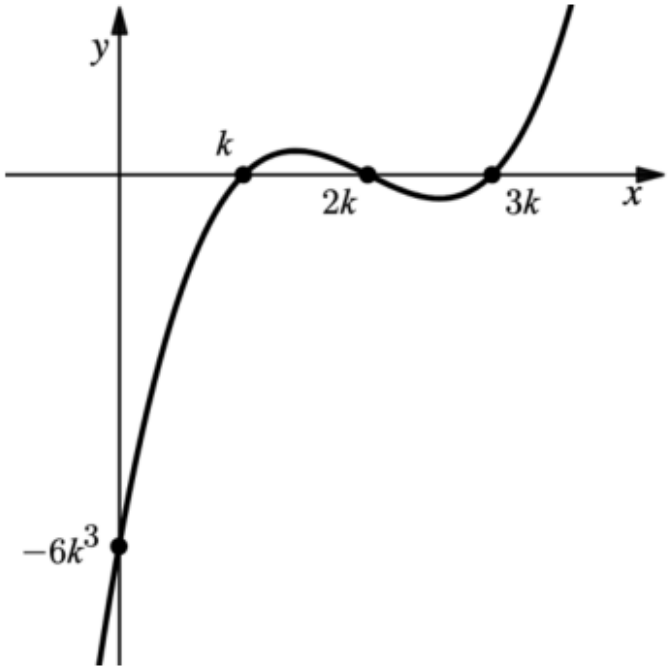
12c)



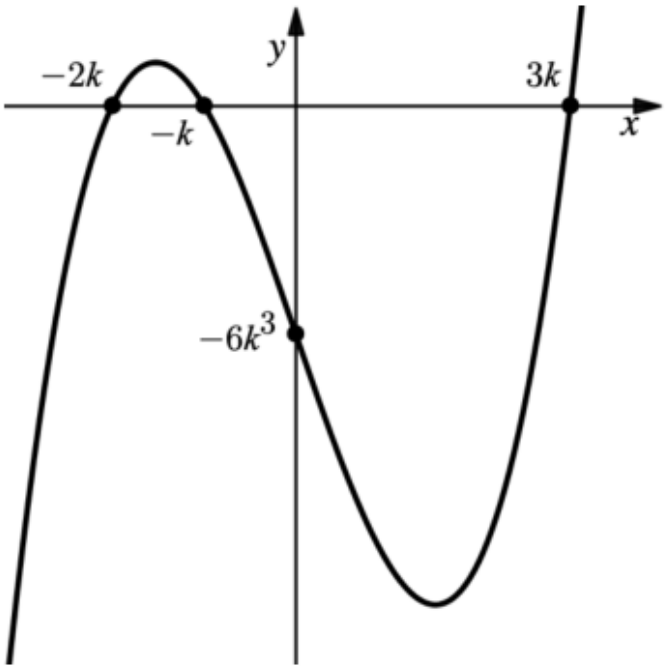
12d)



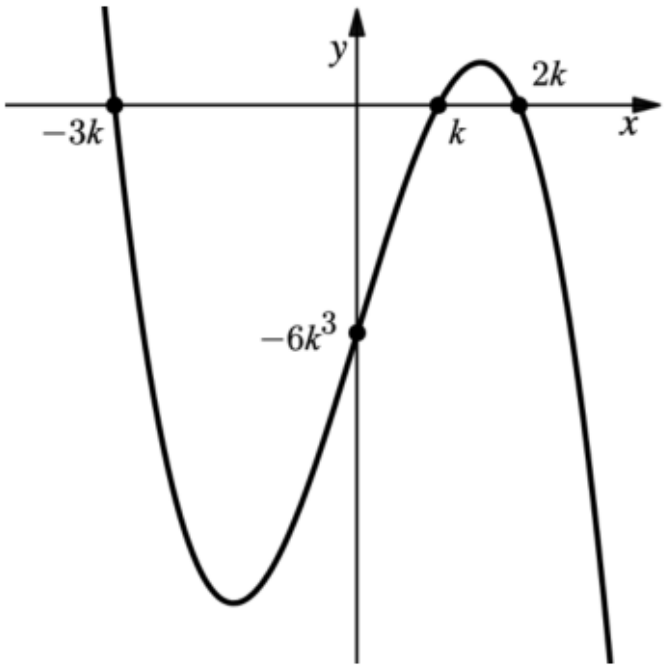
13a)



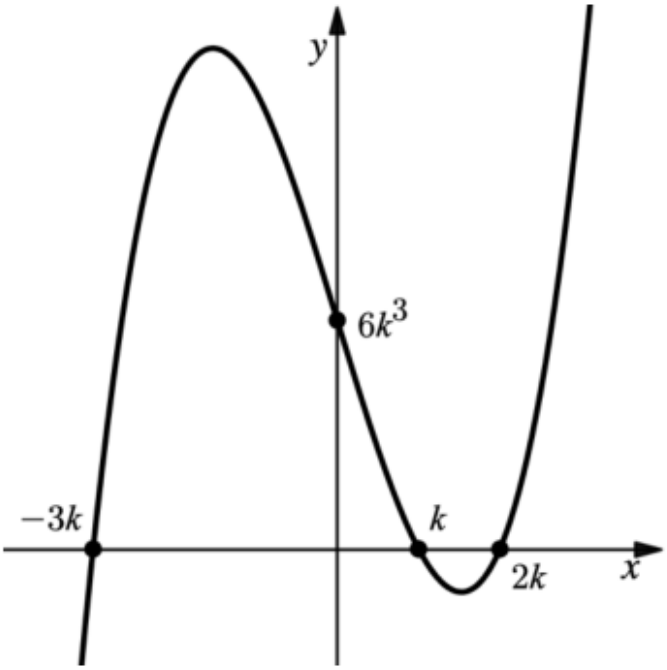
13b)



13c)



13d)



14)

$$y = a\left(x + \frac{5k}{2}\right)\left(x - \frac{6}{k}\right)(x - 6k)$$

$y = 60k$ when $x = 0$:

$$60k = a\left(\frac{5k}{2}\right)\left(-\frac{6}{k}\right)(-6k)$$

$$\Rightarrow 60k = 90ka \Rightarrow a = \frac{2}{3}$$

$$\therefore y = \frac{2}{3}\left(x + \frac{5k}{2}\right)\left(x - \frac{6}{k}\right)(x - 6k)$$

Alternatively:

$$y = a\left(x + \frac{5k}{2}\right)\left(x - \frac{6}{k}\right)(x - 6k)$$

$$\Rightarrow y = \frac{a}{2k}(2x + 5k)(kx - 6)(x - 6k)$$

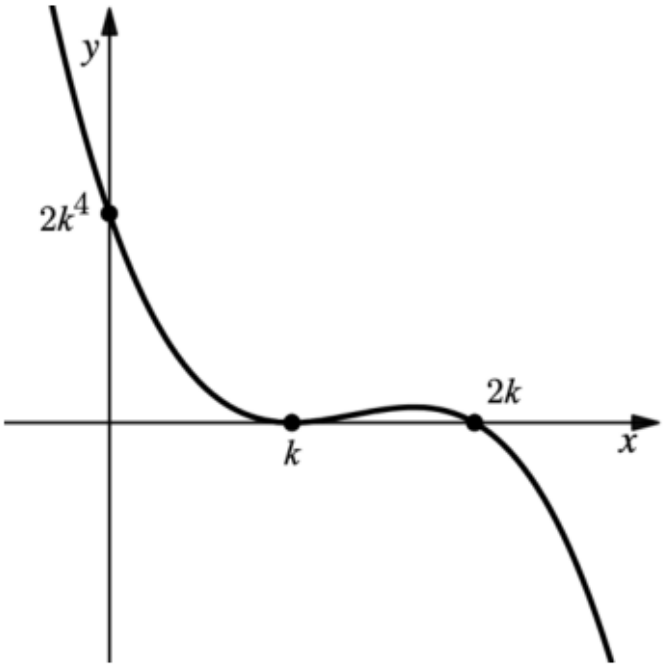
$y = 60k$ when $x = 0$:

$$60k = \frac{a}{2k}(5k)(-6)(-6k)$$

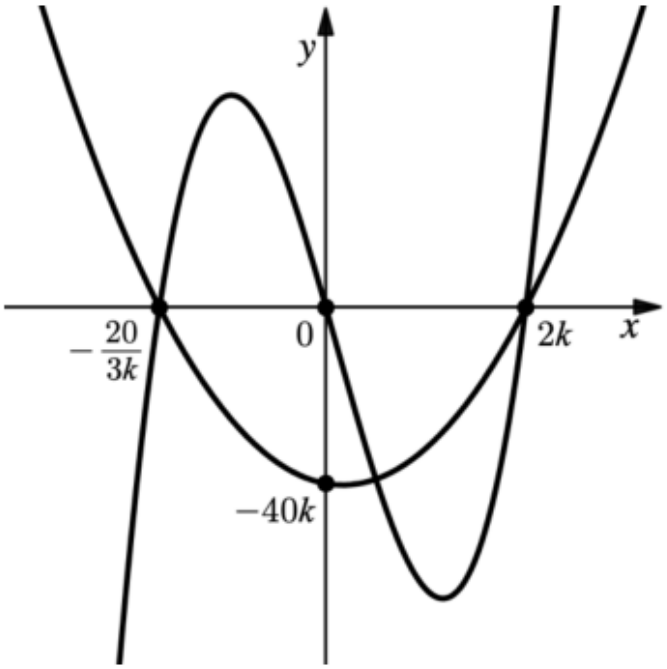
$$\Rightarrow 60k = 90ka \Rightarrow a = \frac{2}{3}$$

$$\therefore y = \frac{1}{3k}(2x + 5k)(kx - 6)(x - 6k)$$

15)



16)



$$(x-2k)(3kx+20) = (x-2k)(3kx+20)x$$
$$\Rightarrow (x-2k)(3kx+20) - (x-2k)(3kx+20)x$$
$$\Rightarrow (x-2k)(3kx+20)(1-x) = 0$$
$$\Rightarrow x = 2k \text{ or } x = -\frac{20}{3k} \text{ or } x = 1$$

When $x = 1$, $y = (1-2k)(3k+20)$

$$\Rightarrow y = -6k^2 - 37k + 20$$

So the curves intersect at $\left(-\frac{20}{3k}, 0\right)$, $(2k, 0)$ and $(1, -6k^2 - 37k + 20)$.